Meowbass

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Meowbass is a psychoacoustic bass enhancer. It splits the bass with a phase-aligned crossover, makes harmonics from it, shapes their level with a dynamics stage and adds them back, so small speakers and headphones suggest deep bass that is not really there. It also lets you pull back the original sub to free headroom.

Quick start

• Set Bass range near the lowest note you care about (kick or bass fundamental, often 50 to 100 Hz).
• Raise Harmonics until the bass feels bigger on small speakers.
• Lower Original bass to free headroom once the harmonics carry the bass.
• Use Even/Odd to choose warm and round (even) or hard and edgy (odd), and Drive to make the harmonics subtle (low) or thick and obvious (high).
• Dynamics (-100 to +100 %): 0 is off. Negative evens out quiet and loud notes (-100 is the most even). Positive makes loud notes get far more harmonics than quiet ones (+100 is drastic). Light or extreme, either side.
• The round gear button in the header opens Settings: the plugin version, a link to the GitHub page and the releases page (it opens your browser; the plugin itself never goes online), and a switch for the info bubbles. Hover any control for about one second to see what it does; the round i button also turns the bubbles on or off.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Frequency

Harmonic filter	Choice: Smooth, Wide, Tight (default Smooth)	Harmonic filter: Smooth is balanced, Wide lets more upper harmonics through (brighter), Tight keeps only the lowest ones (cleaner).
Quality	Choice: Normal / High (2x) (default High (2x))	Quality: High runs the harmonic generator and its filters at twice the sample rate, so hard-driven settings stay clean (less aliasing). It adds no delay and costs very little CPU. Normal is the plain 1x path.
Bass range	30 to 200 Hz default 80 Hz	Bass range: the frequency below which the bass is picked up to make harmonics. Set it near the lowest note you want to feel (kick or bass fundamental, often 50 to 100 Hz).
Original bass	-48 to 6 dB default 0 dB	Original bass: level of the untouched low end. Pull it down to free headroom and let the harmonics carry the bass.
Harmonics	-36 to 18 dB default -12 dB	Harmonics: level of the added overtones that make small speakers suggest deep bass. Start low and raise until the bass feels bigger. Range -36 to +18 dB: from subtle to obvious.

Character

Even/Odd	0 to 1 default 0.5	Even/Odd: character of the harmonics. Even = warm and round (rectifier style), Odd = harder and more edgy (clipper style).
Drive	0 to 100 % default 30 %	Harmonic drive: how hard the bass is pushed into the harmonic generator. Low = clean and subtle, high = thick, obvious and colourful. Works together with Even/Odd.

Dynamics

Dynamics	-100 to 100 % default -40 %	Dynamics: how the harmonics follow the bass level. 0 = off (neutral). Negative = evens out quiet and loud notes (-100 = most even, quiet notes are boosted). Positive = the harmonics follow the level more and more, so loud notes get far more harmonics than quiet ones (+100 = drastic). Tailor it from subtle to extreme either way.
Dyn response	1 to 1000 ms default 40 ms	Dynamics response: how fast the harmonics follow the bass level (works together with Dynamics; at Dynamics 0 it does nothing). Short = every hit snaps the harmonics (punchy, tight); long = the harmonics glide over whole notes (smooth, slow swell). With Dynamics up, the difference is big.

Level

Input	-12 to 12 dB default 0 dB	Input: level into the effect. Raise it for more harmonics from a quiet source.
Output	-12 to 12 dB default 0 dB	Output: final level. Match it with the bypassed level when comparing.

Recommendations

• Check on small speakers or headphones: that is where the effect matters.
• Harmonic filter: Smooth for most material, Wide for brighter results, Tight for the cleanest low harmonics.
• Dyn response (1 to 1000 ms) works with Dynamics (at Dynamics 0 it does nothing). Short values make every hit snap the harmonics (punchy, tight); long values let the harmonics glide over whole notes (smooth, slow swell). With Dynamics up the difference is big.
• Harmonics now range from -36 to +18 dB (default -12 dB): lower for subtle, higher for obvious.
• Version note: the old Dyn ratio knob is replaced by Dynamics (new meaning). Dynamics now runs from -100 to +100 % (0 = off), so an old saved value (0.25 to 20) reads as almost off; set it again; the default Harmonics level is now 6 dB lower than before, so check old projects by ear.
• Do not lower Original bass so far that the club system loses real sub energy.

Presets

Init, Small Speakers, Kick Weight, Sub Clarity, Reese Body, Gentle Warmth, Hardstyle Rumble

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is a screenshot of the real compiled plugin window.



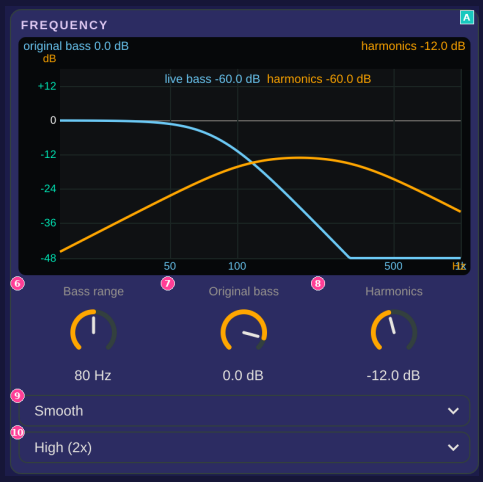
Signal flow

Input > Bass range picks the lows > harmonic generator (Drive, Even/Odd, Harmonic filter) shaped by Dynamics > mixed with the original bass (Original bass and Harmonics levels) > Output.

Header bar

1	**Preset menu**	Factory presets (our own). Use the arrows to step through them.
2	**Randomize**	Set the controls to random values (output level and on/off switches are kept).
3	**Limiter**	Safety limiter on the output: keeps the level from ever passing 0 dBFS, smooth and transparent. Turn it off if you want the raw output (for example to let your own limiter do the work).
4	**Settings**	Settings: version, link to the GitHub page for new versions
5	**Info bubbles (i)**	Info bubbles: hover a control for one second to see what it does. Click to turn them off or on.

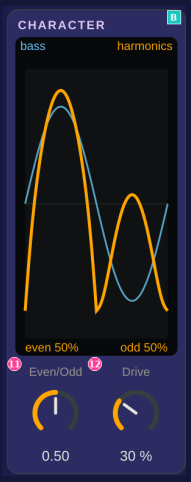
A. Frequency



Display A: Frequency picture. Horizontal: frequency (log), vertical: dB. Blue line: what is left of the original bass (the low-pass set by Bass range). Orange line: the harmonics that are added. The labels give the two levels you set and, under them, the live levels the plugin measures on your signal.

6	**Bass range**	Bass range: the frequency below which the bass is picked up to make harmonics. Set it near the lowest note you want to feel (kick or bass fundamental, often 50 to 100 Hz). (shown: 80 Hz)
7	**Original bass**	Original bass: level of the untouched low end. Pull it down to free headroom and let the harmonics carry the bass. (shown: 0.0 dB)
8	**Harmonics**	Harmonics: level of the added overtones that make small speakers suggest deep bass. Start low and raise until the bass feels bigger. Range -36 to +18 dB: from subtle to obvious. (shown: -12.0 dB)
9	**Harmonic filter**	Harmonic filter: Smooth is balanced, Wide lets more upper harmonics through (brighter), Tight keeps only the lowest ones (cleaner). (shown: Smooth)
10	**Quality**	Quality: High runs the harmonic generator and its filters at twice the sample rate, so hard-driven settings stay clean (less aliasing). It adds no delay and costs very little CPU. Normal is the plain 1x path. (shown: High (2x))

B. Character



Display B: One cycle of a bass wave. Blue is the original sine, orange is the same wave after the harmonic generator. Even % and odd % show the mix of the two harmonic kinds set by Even/Odd.

11	**Even/Odd**	Even/Odd: character of the harmonics. Even = warm and round (rectifier style), Odd = harder and more edgy (clipper style). (shown: 0.50)
12	**Drive**	Harmonic drive: how hard the bass is pushed into the harmonic generator. Low = clean and subtle, high = thick, obvious and colourful. Works together with Even/Odd. (shown: 30 %)

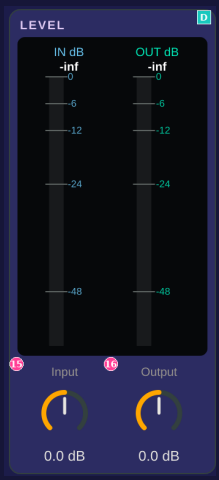
C. Dynamics



Display C: Dynamics law: input bass level (horizontal, dB) against harmonics level (vertical, dB). The thin diagonal is a neutral response, the orange line is the Dynamics setting: above the diagonal on one side of the peak and below it on the other. The white dot is the live level; dyn and the response time are printed in the corner.

13	**Dynamics**	Dynamics: how the harmonics follow the bass level. 0 = off (neutral). Negative = evens out quiet and loud notes (-100 = most even, quiet notes are boosted). Positive = the harmonics follow the level more and more, so loud notes get far more harmonics than quiet ones (+100 = drastic). Tailor it from subtle to extreme either way. (shown: -40 %)
14	**Dyn response**	Dynamics response: how fast the harmonics follow the bass level (works together with Dynamics; at Dynamics 0 it does nothing). Short = every hit snaps the harmonics (punchy, tight); long = the harmonics glide over whole notes (smooth, slow swell). With Dynamics up, the difference is big. (shown: 40 ms)

D. Level



Display D: Input and output meters in dB.

15	**Input**	Input: level into the effect. Raise it for more harmonics from a quiet source. (shown: 0.0 dB)
16	**Output**	Output: final level. Match it with the bypassed level when comparing. (shown: 0.0 dB)

Under the hood

• Even harmonics are warm and round (rectifier style); odd harmonics are harder (clipper style). Even/Odd blends the two.
• Dynamics is bipolar from -100 to +100. 0 is off (neutral). Negative evens out loud and quiet notes (quiet notes get more harmonics); positive makes loud notes carry far more harmonics than quiet ones. Measured on a rumble source, the harmonic level against bass level slope goes from about -0.68 at -100 through 0.40 at off to about 2.0 at +100.
• Dyn response is the follower time (1 to 1000 ms): short snaps on every hit, long glides over whole notes. At Dynamics 0 it does nothing.
• Harmonics are tied to the Bass range band so small speakers suggest deep bass; Original bass can be pulled down to free headroom.
• Knob changes reach the sound within one buffer; smoothing is at most 20 ms. Old saved projects from before the bipolar Dynamics read the new Dynamics as almost off.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.